

Tuhin Chowdhury

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RESEARCH INTERESTS

AI for Software Engineering, Agentic AI, Trustworthy AI.

EDUCATION

BSc in Computer Science & Engineering

Sylhet Engineering College (Affiliated with Shahjalal University of Science & Technology)

Sep 2017 – Apr 2022

Sylhet, Bangladesh

Thesis: “An Automatic System for Identifying and Categorizing Tribal Clothing Based on Convolutional Neural Networks”

CGPA: 3.27 / 4.00 (Final Year GPA: 3.70 / 4.00)

RESEARCH EXPERIENCE

Undergraduate Research Assistant (Thesis)

Machine Learning Lab, Sylhet Engineering College

Nov 2021 – Apr 2022

Sylhet, Bangladesh

- Co-developed a convolutional neural network structure utilizing a custom dataset to categorize traditional patterns, achieving a peak target identification accuracy of 89.97% with YOLOv5 models.
- Curated, preprocessed, and annotated target image metrics to evaluate deep learning frameworks against baseline architectures, establishing updated benchmarking metrics.

PUBLICATIONS & CONFERENCES

Book Chapters

Published

- Chowdhury, T.** et al. (2025). “BLDAR: A Blending Ensemble Learning Approach for Primary Energy Consumption Analysis.” *Springer*.
- Engineered a novel blended ensemble model integrating LightGBM, Decision Tree, AdaBoost, and Random Forest regressors, achieving a state-of-the-art 90% R^2 score.

Conference Papers

Published

- Chowdhury, T.** et al. (2022). “An Automatic System for Identifying and Categorizing Tribal Clothing Based on Convolutional Neural Networks.” *IEEE*.
- Developed a CNN-based processing pipeline utilizing a custom ‘TribalBd’ dataset to classify regional textile variations, achieving an 89.97% accuracy profile via optimized YOLOv5 network models.

Ongoing Research

In Preparation

- Chowdhury, T.** et al. “Retrieval-Augmented Federated Learning: A Survey of Architectures, Security Challenges, Performance Bottlenecks, and Future Directions.”
- Chowdhury, T.** et al. “ActiFed: Active Edge Orchestration in Federated Learning via Dynamic Transmission of Autonomous Logic.”

TECHNICAL SKILLS

Programming Languages: Java 21 (Fluent), Python (Fluent), C++ (Fluent), C (Fluent), C# (Basic)

Application Frameworks: Spring Boot, FastAPI, .NET 8, Flutter, Android

AI & Data Science: MATLAB, PyTorch, TensorFlow, Scikit-Learn, Transformers, Pandas, NumPy, OpenCV, YOLO

Tools & Platforms: AWS EC2, S3, Kubernetes, PostgreSQL, Docker, JUnit, Linux, CI/CD, Git, GitLab, JasperReports, LaTeX

INDUSTRY EXPERIENCE & PROJECTS

Software Engineer

Golden Harvest Infotech Limited

Feb 2023 – Present

Dhaka, Bangladesh

- Led a small backend team to architect and deploy scalable microservices patterns, optimizing automated CI/CD workflows and engineering internal tools supporting 2,000+ daily active users.

- Developed a specialized Nikon SDK-based native hardware integration to establish low-latency, automated image capture and real-time processing operations across the full Agile SDLC.

Project: Hospital Management System (MHM, Houston, TX)

Tech Stack: Spring Boot 3, PostgreSQL, RAG, OpenAI 4, ChromaDB, N8N, AWS S3, Docker

- Engineered robust microservices infrastructures managing cross-department workflows across complex clinical systems (OPD, IPD, IAM, EHR) protected via centralized logging and stateful JWT token authorizations.
- Optimized transactional processing database speeds by implementing advanced indexing matrices, query configurations, and automated asynchronous cron procedures within PostgreSQL.

Project: Church Census Records Archiving Project (Family Search, Utah)

Tech Stack: C#, .NET 8, WPF, GraphQL, SQLite, XAML, Python, Nikon SDK

- Built a high-throughput industrial DSLR imaging system handling 1,000+ legal records hourly, implementing custom computer vision algorithms for automated blur detection and real-time clarity verification.
- Designed an asynchronous concurrent execution pipeline managing synchronized camera triggers, structural image alignment transforms, and multi-frame region-of-interest crops.

Project: Satellite Frequency Renting & Auto Billing System (Bangladesh Satellite Corp., Dhaka)

Tech Stack: Java, Spring Boot, JasperSoft, SendGrid, PostgreSQL, Docker

- Automated 100% of high-volume financial billing workflows, invoice scheduling patterns, automated chalan prints, and graphical reporting features using 15+ complex analytical JasperReports frameworks.

Software Engineer (Intern)

Rokkhi IT Limited

Jun 2022 – Jan 2023

Dhaka, Bangladesh

- Designed an automated optical asset document pipeline by creating a Java service bridged to an EasyOCR Python platform, accelerating data extraction and eliminating manual entry pipelines by over 50%.

ENGLISH TEST

IELTS : 7.0 (Overall) — Listening: 8.0, Reading: 7.0, Speaking: 6.5, Writing: 6.0

Test Date: Dec 2025

AWARDS, FELLOWSHIPS & GRANTS

Selected for Bangladesh-Japan Engineers Training Program

Miyazaki University & Bangladesh Government

2022

Miyazaki, Japan

Best IoT Project Recognition

University Course Project

2019

Sylhet, Bangladesh

First Prize

Intra-Batch Programming Contest

2018

University Campus

First Prize

Essay Writing Competition

2017

University Campus

District Commissioner Scholarship Award

Academic Excellence

2016

Chittagong, Bangladesh

TEACHING & MENTORING EXPERIENCE

Java & Data Structures Instructor

Mentored 10+ students to successful completions

PROFESSIONAL SERVICE

- **Computer Society** – General Secretary Nov 2021 – Nov 2022
- **Bohubrihi (Edtech Platform)** – Campus Director Jan 2021 – Jan 2022
- **SECPA (University Photography Club)** – Vice President & Instructor Jan 2020 – Jan 2022
- **HULT Prize SEC** – Founding Campus Director Jan 2020 – Jan 2021

REFERENCES

Ashraful Islam

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University of Regina, Canada

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Md. Ariful Islam

Chief Technical Officer & System Architect

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